

Kayla McFarlane

kmcfarla@andrew.cmu.edu | 631-377-2267 | [linkedin.com/in/kayla-mcfarlane28](https://www.linkedin.com/in/kayla-mcfarlane28)

EDUCATION

Carnegie Mellon University

Master of Science in Robotics

Bachelor of Science in Electrical and Computer Engineering (*Software Systems Concentration*)

Pittsburgh, PA

August 2027

May 2025

RELEVANT COURSEWORK

Computer Systems, Computer Security, Computer Vision, Applied Cryptography, Data-Focused Python, Mobile Web Design & Development, Web Application Development, Human-Robot Interaction, Ethics and Robotics, Mobile Robots, Machine Learning, Math Fundamentals for Robotics, Embodied AI Safety

AWARDS & HONORS

Rales Fellow, Carnegie Mellon University

2025 – 2027

- Awarded a prestigious full-tuition fellowship to one of 34 underrepresented students in STEM from 400+ applications, providing mentorship, career coaching, and professional development opportunities to cultivate future STEM leaders.

ETH Zurich Robotics Summer School, Selected Participant

June 2025

- One of 50 students globally selected from 292 applicants for a fully funded summer program on robotic optimization and field deployment in real-world environments.
- Includes hands-on UGV programming, Search and Rescue challenges, and exposure to real-world robotic systems in disaster environments

RESEARCH EXPERIENCE

Graduate Research Assistant

May 2025 – Present

SAFE Project & FireTech Initiative, CMU AirLab

- **Leading** development of benchmark to assess the **safety and performance** of navigation algorithms under **uncertainty in complex, real-world environments**.
- Conducted extensive literature reviews on safety frameworks for autonomous robots, analyzing approaches from collision avoidance to perception-aware navigation
- Designing test scenarios that systematically vary key environmental factors (i.e sensor range, obstacle density, perception quality, and external forces) to challenge and compare navigation algorithms
- Developing Python scripts in Isaac Sim to generate randomized testing environments with varied hazards for evaluating navigation algorithm performance across different robot embodiments
- Developing new metrics that assess, not just task completion, but the *quality of decision-making under uncertainty*, incorporating concepts of risk and behavioral safety
- Collaborating on testing protocols to evaluate how robust and adaptable navigation algorithms are under real-world mission constraints and degraded conditions
- Authored workshop paper on multi-modal failure adaptation, accepted to IROS 2025 Workshop

Undergraduate Research Assistant

Mar 2024 – May 2025

SAFE Project, CMU AirLab

- Contributed to the **SAFE (Safe Agile Foresight Everywhere) project**, focused on enabling autonomous drone navigation in unknown, dynamic, and cluttered environments with degraded perception
- Conducted extensive literature reviews on aerial robotics safety frameworks, analyzing approaches from collision avoidance to perception-aware navigation
- Collaborated with graduate students to systematically evaluate state-of-the-art methods for agile flight, identifying key limitations and opportunities for improvement
- Participated in weekly research discussions to develop novel approaches combining neural Control Barrier Functions (CBFs), Hamilton-Jacobi reachability, and Vision-Language Models
- Set up and configured 3D simulation environments, in **Isaac Gym**, for testing perception-conditioned Neural CBFs, maintaining comprehensive documentation
- Contributed to research on **Perception-Aware Neural Control Barrier Functions** for adaptive safety in degraded visual environments

Undergraduate Research Assistant

Jan 2023 – Feb 2024

Wildfire Project, CMU AirLab

- Contributed to the **Wildfire Project**, aimed at providing real-time situational awareness to firefighters during wildfires.
- Worked on **FIReVision**, a photorealistic wildfire simulation tool designed to test drone navigation, object detection, and sensor integration.
- Designed and implemented automated data collection pipeline for FIReVision, generating **15,000+** multimodal training images (RGB, thermal, depth, segmentation)
- Built modular **Python** scripts for drone navigation control, object detection, and systematic synthetic data gathering in FIReVision environments
- Created comprehensive testing framework to validate and debug pipeline components, ensuring data quality for neural network training
- Created realistic smoke simulation within FIReVision, using Unreal Engine and FluidNinja for testing drone perception in degraded visual conditions
- Contributed to documentation and demo materials for Epic Games MegaGrant application, showcasing simulator capabilities
- Implemented interactive firefighter character in WUI simulation for stakeholder demonstrations at 2024 Wildfire Symposium, enabling hands-on testing of the system
- Presented research at CMU Meeting of the Minds Symposium

Human-Robot Interaction Research Project

Nov 2023 – Dec 2023

Human-Robot Interaction, Carnegie Mellon University

Exploring Accent Variability, Phrase Difficulty, and Speaker Distance in Robot-Enabled Speech Recognition

- Evaluated Google Speech Recognition integrated with Pepper robot across diverse accents (American, Chinese, Guyanese, Brazilian) to assess ASR performance in classroom settings
- Developed Python modules for real-time transcription and analyzed performance using Translation Error Rate (TER) metrics
- Identified critical limitations in ASR accuracy for accented speech and distance variability, providing recommendations for improving accessibility in educational settings
- Project code adopted by AI Maker Space for demonstrating speech recognition challenges

Undergraduate Research Observer

Nov 2024

The MindfulNest Project, CMU CREATE Lab

- Observed real-world interactions between neurodivergent children and **MindfulNest**, a socially assistive emotional regulation robot
- Documented and **analyzed behavioral patterns** to identify design insights around engagement, comfort, and emotional expression
- Supported data collection and contributed insights to refine the pilot study methodology

INDUSTRY EXPERIENCE

Software Engineering Intern

Jun 2024 – Aug 2024

DigITs Program, Memorial Sloan Kettering Cancer Center (MSK)

- Developed AI chatbot interface using React and TypeScript for patient portal serving **170,000+** users annually
- Streamlined LLM fine-tuning process for **50+** developer teams, improving AI integration across departments
- Studied full-stack architecture of MyMSK's Self-Scheduling system (**React** frontend, **C#.NET** backend, **SQL** Server).
- Led development of MSK SkinMonitor, winning 1st place in intern hackathon for AI-powered health tracking app
- Practiced Agile methodologies through daily stand-ups and two-week sprint cycles

TECHNICAL PROJECTS

INFERNOT – AI-Powered Wildfire Defense System

November 2025

CMU Hack-A-StartUp Competition

- Collaborated with 4-member team to develop automated wildfire protection system using AI sensors and robotics
- Led robotics integration strategy for automated deployment system using thermal imaging, computer vision, and air-quality sensing
- Contributed to technical feasibility analysis and prototype development planning
- Placed 4th out of 15 teams in university-wide entrepreneurship competition

Nightingale – Voice Cleaner and Transcriber

October 2025

Wireless Innovation Hackathon for Accessibility

- Developed web-based assistive technology to enhance voice clarity and communication for individuals with vocal cord disorders in noisy environments
- Implemented audio processing pipeline using Sox for noise reduction and Whisper AI for real-time speech transcription
- Designed future wearable device concept (Nightingale Clip) for portable, real-time voice assistance
- Collaborated with interdisciplinary team and engaged directly with disability community stakeholders

Study Bearbot – Interactive Desktop Companion for Focus and Stress Relief

Jan – May 2025

Electrical and Computer Engineering Capstone Project, Carnegie Mellon University

- Collaborated with a partner to design and build a **multisensory desktop robot** that supports student focus and stress reduction
- Led firmware development on the **Arduino Nano RP2040**, integrating tactile button inputs, toggle switches, rotary encoder, and OLED display to control LED lighting, ultrasonic diffuser, and audio playback with sub-500ms responsiveness
- Built a **React**-based web app for real-time, two-way synchronization with the robot, enabling remote control of sensory features (LED, aroma, audio), timer tracking, and dynamic motivational quote display through HTTP communication
- Achieved an average **7.8-hour** battery life and demonstrated **measurable reductions in user stress**
- Conducted an **8-user study** demonstrating reduced stress and increased focus, integrating survey data and interaction logs to inform design iterations

Vivian: A Caribbean Voice Assistant

Feb 2024

SheInnovates Hackathon 2024

- Developed voice assistant prototype for underrepresented English dialects using OpenAI Whisper
- Implemented speech-to-text system for Caribbean Patois using Python and custom API integrations
- Led technical design and development of speech processing pipeline
- Addressed challenges in training a potential AI model due to insufficient datasets for Caribbean accents, highlighting the need for further data collection and model fine-tuning for underrepresented English dialects

PUBLICATIONS

- Navin Sriram Ravie, Andrew Jong, **Kayla McFarlane**, Jay Patrikar, Bijo Sebastian, Sebastian Scherer. "Don't Fool Me Twice: Adapting to Failures in the Wild with Multi-Modal Reasoning." *IROS 2025 Workshop on The Art of Robustness: Surviving Failures in Robotics*, October 2025.

VOLUNTEER

Conference Volunteer

Jan 2026

IEEE/SICE International Symposium on System Integration (SII 2026) | Cancun, Mexico

- Supported conference operations for international robotics and systems integration symposium
- Assisted with session coordination and attendee support

Outreach Committee Member

Oct 2024 – Dec 2024

Society of Women Engineers (SWE), Carnegie Mellon University

- Participated in biweekly planning meetings for SWE's High School Day, contributing to event logistics and programming
- Helped coordinate and set up the event day activities and venues
- Introduced young women to engineering through hands-on experiences and provided mentorship during the event

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, C#, HTML/CSS, SQL, MATLAB

Frameworks/Libraries: React, .NET Core, Node.js, Express.js, Django, Bootstrap

Tools: Git, GitHub, Linux, Docker, Figma, Isaac Gym/Sim/Lab, Gazebo, ROS2

Technologies: SQL Server, REST APIs, Agile/Scrum

Technical Areas: Full-Stack Development, Computer Vision, Machine Learning/AI, Embedded Systems, Real-time Systems, API Integration, Sensor Integration

EXTRACURRICULARS

Women in Electrical and Computer Engineering (WinECE), Society of Women Engineers (SWE), Carnegie AI Safety Initiative (CASI), Paths to AI Research (PAIR)